ETL Process Summary – Nova Scotia Air Quality Project

Extract

Air-quality data was obtained from the **Government of Canada Open Data Portal**: https://search.open.canada.ca/opendata/?search_text=air+quality

Five CSV files related to ambient air-quality monitoring stations in Nova Scotia were downloaded. Each file represents observations for specific pollutants and locations, with historical coverage varying by station and pollutant. The raw datasets span from **1995 to 2025**, depending on availability. Due to file size constraints, the raw data is documented rather than fully stored in the repository.

Transform

Data transformation and cleaning were performed using **Python (pandas)** and documented in a Jupyter Notebook. Each raw file was cleaned individually to remove null and invalid values, standardize column names, and ensure consistent formatting. After cleaning, all datasets were appended into a single **staging table**, creating a unified and structured dataset for analysis.

From the staging table, the data was scoped to the **2015–2025** period to align with analytical requirements. Time-related attributes were derived to support analysis. Although SO_2 data exists in the staging layer, it was excluded from the final dataset because no SO_2 observations were available within the selected time window.

Load

The final clean dataset contains validated daily air-quality records for **O_3 (Ozone)** and **NO_2 (Nitrogen Dioxide)** across selected monitoring stations in Nova Scotia. This dataset serves as the primary analytical layer and is loaded into **Power BI** for AQI-focused analysis, seasonal trend exploration, and visualization.